# Equity by Design: A Distribution-Focused PISA Data Dashboard

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## Executive Summary

Educational assessment data is frequently reported as static average scores, omitting crucial context and leading policymakers to attribute performance disparities to demographic groups rather than systemic inequities. To address this, we developed an interactive data product for the Educational Testing Service (ETS) that visualizes full score distributions and contextual variables using Programme for International Student Assessment (PISA) data.

The dashboard draws on student records from the 2015, 2018, and 2022 PISA cycles. Rather than reporting country averages alone, it surfaces the full score distribution, broken down by percentile, socioeconomic background, immigration status, school type, and location. A Parquet and DuckDB backend keeps query times low even at this scale, by pushing filters and aggregations down to the data layer before results reach the application.

The tool offers two modes: an open Explore view for analysts who want to slice the data themselves, and a guided Story mode that walks policymakers through key patterns with contextual framing. Sampling weights and plausible values are applied throughout, following PISA's methodology. The goal is to make it easier to ask more specific questions, not just how a country scores on average, but how different groups of students are distributed across the full range of performance.

## Introduction

### Background and Problem Statement

International educational assessments serve as a primary benchmark for governments evaluating the efficacy of their education systems. However, relying solely on average scores to compare student groups obscures critical information: it hides within-group variation and ignores structural inequities, such as disparities in school resources.

Research suggests that mean-centric reporting leads audiences to attribute score differences to intrinsic group traits rather than systemic factors (Holder and Xiong 2022), which can shape how education departments allocate resources. When data is presented without distributional context, students who need the support are at risk of being mischaracterized. Aligning with ETS's commitment to fairness and equity in assessment (Educational Testing Service 2026), this project explores how shifting the way data is presented can open up more grounded, contextually aware policy conversations.

### Refined Scientific Objectives

While the initial goal was to build a dashboard to counter deficit thinking, it shaped three concrete objectives that guided the technical and design work:

- **Objective 1: Handling complex survey data at scale.** PISA data requires computing scores across 10 plausible values per subject and applying replicate weights to produce accurate population estimates. The objective was to build a data pipeline that could handle these calculations dynamically across three survey cycles, without making users wait.
- **Objective 2: Showing distributions, not just averages.** Rather than summarizing performance as a single number, the visualizations show where students are actually spread across the score range. A shaded percentile distribution plot displays each group's full distribution, with darker shading where students are most concentrated and a median marker for reference. The same format is used consistently across the score profile view and group comparisons, so patterns are easy to read across different breakdowns. A separate chart tracks how scores at different percentiles have shifted over PISA cycles, making trends visible beyond the average.
- **Objective 3: Supporting different kinds of users.** The dashboard offers two entry points. The Explore mode lets analysts and researchers interact with the data directly, cross-referencing variables like socioeconomic background, immigration status, and school type without a fixed path. The Story mode takes a different approach, guiding policymakers through key findings with contextual framing built in. Automated text generation accompanies each chart, describing patterns in neutral language and avoiding interpretations that reduce group differences to inherent traits.

## Data

### How the data was collected

The dataset originates from the Programme for International Student Assessment (PISA), administered by the OECD every three years to nationally representative samples of 15-year-old students (OECD 2023). PISA combines a cognitive assessment in mathematics, reading, and science with background questionnaires completed by students, school principals, and (optionally) parents, covering socioeconomic status, immigration background, school characteristics, and attitudinal measures such as sense of belonging. Countries administer the test under standardized international protocols, with national sampling frames stratified by school type and region to ensure representativeness within each participating system. The dashboard draws on the three most recent cycles - 2015, 2018, and 2022, covering 98 countries and economies, including all OECD member states and a set of non-member "partner" economies.

### Observational units

The unit of observation is the individual student. Each record corresponds to one student's responses to the assessment and contextual questionnaires, linked to school-level identifiers (`CNTSCHID`) and a final sampling weight (`W_FSTUWT`) used to produce nationally representative estimates. Because PISA does not report a single observed score per student, the target variables are 10 plausible values per subject (`PV1MATH`–`PV10MATH`, and equivalently for reading and science) - multiple imputed draws from the posterior distribution of each student's latent ability, designed to account for measurement uncertainty in a low-stimulus assessment (OECD 2024). All score statistics reported in the dashboard (means, percentiles, and standard errors) are computed by averaging across all 10 plausible values per the PISA technical standards (OECD 2024), rather than treating any single plausible value as a point estimate.

#### Size and characteristics of the dataset

Across the three cycles, the combined dataset comprises 1,745,082 student records and approximately 1,700 raw columns per cycle. After removing columns with no analytic use to this project (entirely empty optional questionnaire fields, redundant identifiers), the working column set was reduced to 133 variables per record, covering the 30 plausible value columns, sampling and replicate weights, and a core set of equity-relevant context variables (socioeconomic status, immigration background, gender, school location and type, language spoken at home, and grade repetition). Sample sizes vary substantially by country: Canada, for instance, contributes 23,073 student records in our working sample, while the United States contributes 4,552 - a roughly 5.1-fold difference that makes correct application of sampling weights essential to any cross-country comparison, rather than an optional refinement.

#### Data quality issues, missingness, biases or limitations

Missingness was assessed across the core contextual variables used in the dashboard's equity breakdowns, summarised in Table 1. Immigration background (`IMMIG`) is the most affected of the retained variables. Columns drawn from the optional parent and student wellbeing questionnaires were 100% missing and were dropped entirely. A second limitation concerns comparability across cycles: the OECD periodically revises questionnaire item wording and response categories between PISA administrations, meaning a small number of contextual variables are not perfectly comparable across 2015, 2018, and 2022. The dashboard surfaces this to users directly through warning messages rather than silently interpolating or assuming equivalence.

Table 1: Missingness in key contextual variables

| | Variable | Missing (%) |
|---|---|---|
| 0 | Gender (ST004D01T) | 0.0% |
| 1 | School type (SC001Q01TA) | 6.5% |
| 2 | ESCS index | 3.2% |
| 3 | Immigration background (IMMIG) | 6.1% |

### How the available data shaped or constrained the scientific objectives

The structure of the PISA data directly constrained what the dashboard could responsibly claim to show. The use of plausible values rather than single test scores meant that every statistic had to be computed as an average across ten imputations, ruling out any shortcut that treated `PV1` as a representative score. The replicate weight structure (80 Fay BRR weights per student) (OECD 2024) made standard error estimation considerably more involved than a naive weighted standard deviation, but was necessary to flag country-year-subject combinations where small sample sizes made point estimates unreliable, which directly shaped the decision to add standard error and 95% CI to the dashboard's summary statistics. Finally, because PISA microdata cannot support causal claims, the project's scope was deliberately limited to descriptive and comparative reporting (surfacing the full distribution of outcomes and how equity-relevant gaps move over time) rather than attempting to model or explain why those gaps exist, which would have required a different kind of data than PISA's cross-sectional design provides.

## Data Science Methods

### Key exploratory findings

Two findings from exploratory analysis directly motivated the dashboard's design. First, aggregate score changes obscure highly uneven shifts across the distribution (Malkus 2023): in Canada, for example, mathematics scores declined by 16.6 points at the 10th percentile between 2018 and 2022, much more than the 10.2-point decline observed at the 90th percentile. A single national average masks this - it would report a uniform decline where none occurred (Figure 1).

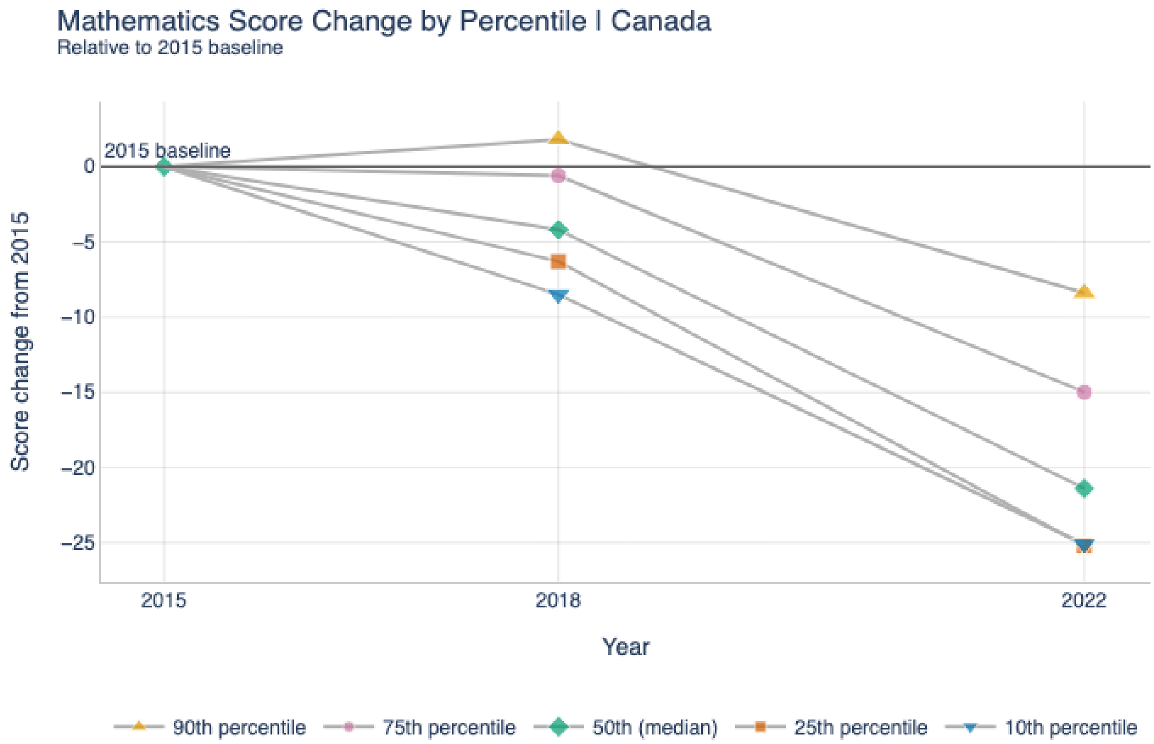


Figure 1: Mathematics score change by percentile relative to 2015 baseline | Canada

Second, within-group variation is often as large as between-group variation: students attending school in villages or large cities show overlapping score ranges rather than cleanly separated distributions, meaning that geographic or socioeconomic group membership alone is a poor predictor of any individual student's outcome (Figure 2).

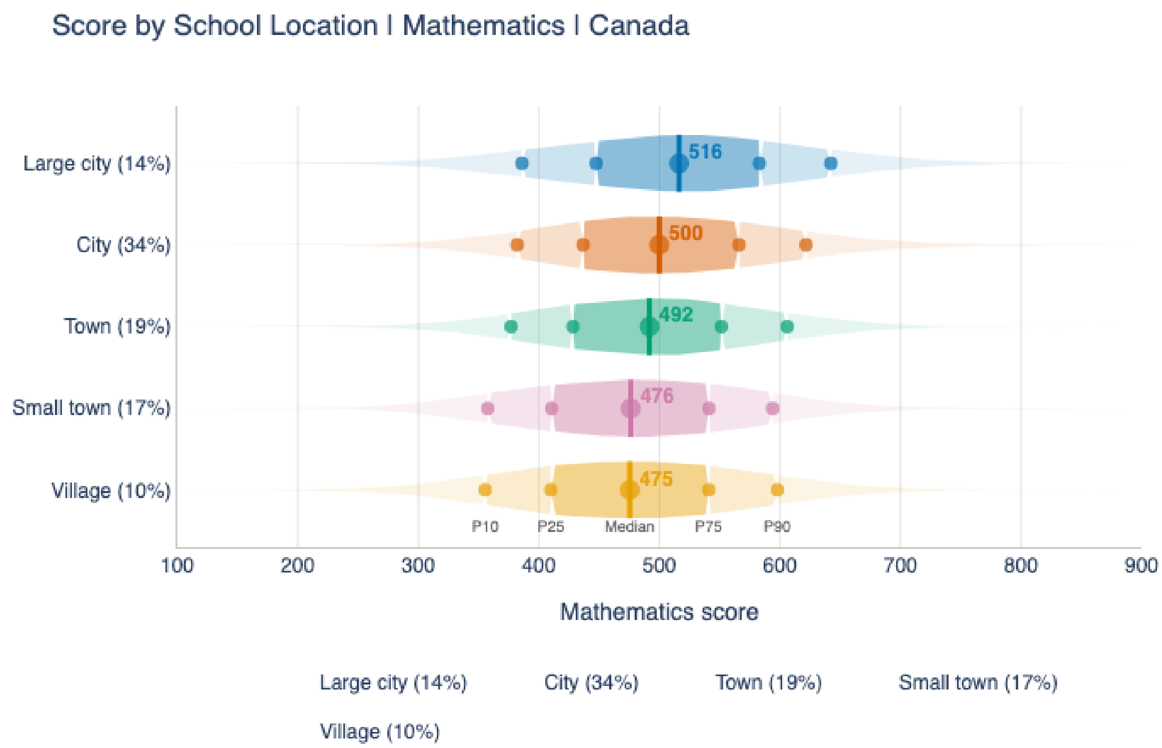


Figure 2: Mathematics score distribution by school location | Canada, 2022

### Data pre-processing pipeline

Several preprocessing steps were necessary before the data could be used in the dashboard. Student and school-level files were merged on the school identifier for each cycle, then written to per-year Parquet files rather than kept in CSV, since the full multi-cycle dataset could not be loaded into memory as a single pandas DataFrame without causing out-of-memory failures in the deployed environment. Querying is instead performed on demand against the Parquet files using DuckDB, so that any given chart only loads the specific columns and country-year combinations it needs rather than the full 1.7 million-row dataset. Records missing the final sampling weight were excluded from all weighted calculations, since an unweighted estimate would not generalize to the national population PISA is designed to represent. Subgroup breakdowns (by gender, immigration status, socioeconomic quartile, school type, and school location) are suppressed in the dashboard whenever the underlying cell falls below 30 valid observations, to avoid presenting unstable percentile or mean estimates as if they were reliable.

Two additional pre-aggregation steps are run at pipeline build time to keep render times low. Gaussian KDE curves and weighted percentiles (P10-P90) are pre-computed across every combination of country, year, subject, subgroup, and stored in `pisa_precomputed.parquet`, so density charts render from a lookup rather than fitting KDEs at runtime. Standard errors are similarly pre-aggregated into `pisa_ se_stats.parquet`, indexed by country, subject, and year using the 80 replicate weights averaged across all 10 plausible values, allowing constant-time lookup instead of loading 91 columns per chart render.

### Approach

We used a descriptive and distributional analytics approach rather than a predictive modeling approach. This decision was driven by the partner's need: the goal was not to predict an individual student's achievement, but to help stakeholders explore educational equity patterns in PISA data and understand how performance differences vary across countries, subjects, and student subgroups. Therefore, the main task was to transform complex assessment data into statistically meaningful and interpretable visual summaries.

The dashboard focuses on both aggregate and distributional summaries. Weighted mean and median scores are useful for identifying whether a gap exists, such as whether students from higher socioeconomic status groups tend to outperform students from lower socioeconomic status groups. However, measures of central tendencies compress the distribution into a single number. For an equity-focused question, the more useful follow-up is often where the gap occurs. For this reason, we computed weighted percentile profiles across the score distribution. These views allow users to examine whether subgroup gaps are concentrated among lower-performing students, higher-performing students, or remain consistent across the distribution.

Several subgroup variables were selected because of their relevance to equity-oriented education questions: socioeconomic status, gender, immigration status, school location, and school type. For socioeconomic status, students were divided into quartiles based on the ESCS (socioeconomic) index. This made the comparison interpretable and allowed us to contrast students in the lowest and highest socioeconomic groups.

### Visualization Choices

We build two modes in the dashboard to support two different user needs. In the story mode, users can choose the country and subject on the side bar and get a detailed report, with questions to guide them through which plots to look at, how to interpret and the most important findings from these plots. In the exploration mode, users can interactively change the country, subject, year, and subgroup to investigate patterns in more detail. Percentile or distribution plots, heatmaps, and scatterplots are useful in this mode because they show the distribution and spread of the data, preserve more information and allow flexible comparison.

### Limitations and Ethical Considerations

The main limitation of these methods is that the dashboard provides descriptive evidence rather than causal estimates. Differences between groups should not be interpreted as the effect of socioeconomic status, immigration status, school type, or any other factor. These variables are correlated with many contextual and institutional factors that are not fully adjusted for in the dashboard. This evidence of difference should be seen as a guide to further investigation on why students in certain groups have a lower performance and what help would be useful, instead of showing certain groups as deficient. As a result, the dashboard is best understood as an exploratory decision-support tool rather than a formal inferential modeling system.

The main stakeholders affected by these methodological decisions are education ministry staff, policy analysts, and other users preparing PISA-based policy briefs. If the dashboard overemphasized rankings, minor differences or single averages, users might draw overly simplified conclusions about national performance. By showing distributions, subgroup gaps, and uncertainty warnings, the dashboard encourages more careful interpretation. This matters because policy decisions based on education data can influence how attention and resources are directed across student groups, schools, or regions.

There are also ethical considerations in how the dashboard presents subgroup comparisons. Variables such as immigration status, socioeconomic status, gender, and school type can be sensitive if presented as explanations for performance differences. The dashboard therefore treats these variables as descriptive comparison groups rather than causal drivers. It can show that two groups differ in score distributions, but it cannot determine whether the group characteristic caused the difference. Observed gaps may reflect many overlapping factors, such as socioeconomic sorting, language background, school resources, or regional policy context.

## Data product and results

### Product Description

The PISA Smart Dashboard is a web application built using Streamlit and Plotly, aimed at education ministry staff who work with PISA data when preparing policy briefs. The dashboard has two main modes. The Data Story tab guides users through a four-chapter narrative with a summary of the key findings at the end, for a selected country and subject. The Explore tab lets users freely choose chart types, countries, and years without a fixed structure.

A key motivation behind the Data Story tab was feedback from ETS that most policymakers look only at mean scores, which hides how spread out student performance actually is within a country, and also hides differences between student groups. To address this, the narrative moves through four chapters: where the country stands relative to the OECD average, how scores have changed across PISA cycles, how outcomes differ by socioeconomic background, immigration status, and gender, and finally how school location and type relate to performance. Each chapter includes an automatically generated finding based on the computed data, alongside a plain-language policy note. The Explore tab adds chart types like a country-level scatterplot, an intersectional heatmap, and a side-by-side comparison view for users who want to dig deeper.

### Why This Form

We chose Streamlit over alternatives like Dash or a React frontend mainly because the tool is intended for internal use by ETS and ministry staff rather than the general public. Keeping everything in Python meant the data pipeline, statistical functions, and UI all lived in one codebase, which made it much easier to respond to partner feedback quickly across the project timeline.

The focus on full score distributions rather than averages came from looking at the data directly. The spread from the 10th to the 90th percentile within a typical country is around 240 points, while the gap between an average country and the OECD mean is closer to 30 points. Showing only averages would draw attention to the smaller number. Shaded density charts and percentile-based views were used instead of simple line plots to make the within-country spread readable to non-statisticians.

The OECD average is used as the reference point throughout the dashboard, which was a specific request from our ETS partner. We also decided not to automate peer country comparisons, since suggesting which countries to compare against would risk encoding assumptions we cannot justify given the diversity of education systems in the dataset.

### Strengths and Limitations

On the statistical side, the dashboard follows PISA methodology correctly. Weighted percentiles are averaged across all 10 plausible values (OECD 2024) using `DescrStatsW` with the `W_FSTUWT` sampling weight. Standard errors account for both sampling variance from the 80 Fay BRR replicate weights and imputation variance across plausible values (OECD 2024).

The two-level callout system (blue for key findings, green for policy implications) came directly from ETS feedback. Earlier versions stacked colored boxes on every chapter, which felt overwhelming. The added key insight only shows up when the data actually warrants extra attention, for example when the SES gap within a country is larger than the gap between that country and the OECD average.

The dashboard currently covers 98 countries across the 2015, 2018, and 2022 PISA cycles and three subjects. Data is stored as Parquet files on S3 and queried with DuckDB, so only the columns and rows needed for a given chart are loaded into memory. A small pre-aggregated summary file is used to populate the sidebar without touching the full dataset.

There are some limitations worth noting. The app is deployed on Posit Cloud, which imposes memory constraints that the pipeline's pre-aggregation steps were specifically designed to work around, with DuckDB handling all filtered queries at runtime. Cold starts remain slow regardless, since Streamlit reinitializes the session on each new connection. The processed parquet files are currently hosted on an S3 bucket tied to an AWS account with a set validity; after that date, any deployment relying on it will fail to load data. Moving the 4 parquet files to a more permanent hosting location would be a necessary step before the dashboard can be used reliably beyond that date.

### Future Directions

The data pipeline is already set up to handle additional PISA cycles, so adding 2009 and 2012 data would not require major changes. Longer term, expanding country coverage beyond the current 98 and adding earlier cycles would strengthen the trend chapter by giving it more data points to work with.

## Conclusions and Recommendations

Education ministries tend to focus on a country's mean PISA score and where it ranks internationally. While this is a natural starting point, it misses two things that are arguably more useful for domestic policy: how spread out student outcomes are within the country, and which groups of students are falling behind. The dashboard was built to shift that focus, putting the full score distribution front and center and making equity breakdowns easy to access alongside the international comparison.

Overall the product does what it set out to do. A user can open the Data Story tab for any country, pick a subject, and within a few minutes see how their country compares to the OECD average, whether scores have gone up or down across cycles, which student groups have the largest score differences, and how urban and rural or public and private schools compare. The auto-generated text on each chart means users do not need to interpret the visuals from scratch. Using the OECD average as the single reference point throughout keeps comparisons consistent and avoids the bias that would come from hand-picking peer countries.

There is also a deeper limitation that is harder to solve. The dashboard can show that first-generation immigrant students score 40 points below native students in a given country, but it cannot tell the user why. Whether that reflects a lack of language support, socioeconomic sorting, geographic concentration, or something else entirely depends on the context that lives outside the PISA dataset. The tool is best thought of as a starting point for analysis rather than a final answer.

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